

---

# Regime-Aware Adaptive Conformal Prediction for Time Series via Hidden Markov Models

---

Anonymous Author  
Anonymous Institution

## Abstract

Adaptive conformal prediction (ACP) produces distribution-free prediction intervals for time series by adjusting the target miscoverage level online. However, ACP calibrates on a single sliding window of nonconformity scores: when the data alternate between regimes of different volatility, large scores collected in turbulent periods contaminate calibration in calm periods, yielding needlessly wide intervals, and vice versa. We propose a regime-aware extension of ACP that couples it with a hidden Markov model (HMM) fitted to log-scaled forecasting errors. The method maintains a separate, posterior-weighted calibration set and a separate miscoverage level for every hidden state; at prediction time the interval is constructed from a weighted empirical quantile of the calibration set associated with the most likely state. We prove that the method inherits the asymptotic coverage guarantee of ACP, both marginally and within every regime. On a synthetic regime-switching benchmark and two real data sets—household electric power consumption and ECG recordings—the proposed method stays within 0.01 of nominal coverage at every target level while reducing median interval width by up to 71% and mean width by up to 34% relative to ACP. In the same settings, the recent baselines EnbPI and SPCI violate nominal coverage by up to 0.034 and 0.113, respectively.

## 1 INTRODUCTION

Point forecasts are rarely sufficient in decision-critical applications of time series analysis. In energy markets, bidding strategies, storage arbitrage, and balancing operations require price forecasts with quantified uncertainty rather than a single expected value (O’Connor et al., 2025).

In manufacturing, conformal anomaly detection over sensor streams enables predictive maintenance with a statistically guaranteed false-alarm rate, in contrast to fixed-threshold rules (Yuan et al., 2026). In logistics, predictive distributions over order fulfillment times support reliable delivery promises and warehouse allocation (Ye et al., 2025). Conformal prediction (CP) is an attractive tool in all these settings because it wraps around any forecasting model and produces prediction intervals with a prescribed coverage level  $1 - \alpha$ , without distributional assumptions beyond exchangeability (Vovk et al., 2005; Angelopoulos and Bates, 2023).

**The problem.** Exchangeability is systematically violated by time series: observations are serially dependent and their distribution drifts. Adaptive conformal inference (ACP; Gibbs and Candès, 2021) restores validity by updating the working miscoverage level online and by calibrating on a sliding window of recent nonconformity scores. Yet ACP retains a *single* calibration set. Real-world series switch between regimes in which the forecasting model performs very differently—calm versus volatile periods, day versus night load, normal versus arrhythmic heartbeats. After the series leaves a turbulent regime, the large scores accumulated there keep dominating the shared calibration window, so intervals remain wide long after the disturbance has passed (Figure 1); symmetrically, calm-period scores make intervals too narrow when volatility returns, and coverage must be repaired slowly through the  $\alpha_t$  recursion.

**Our approach.** We make ACP *regime-aware*. A hidden Markov model (HMM) is fitted to the log-scaled forecasting errors of the point predictor and acts

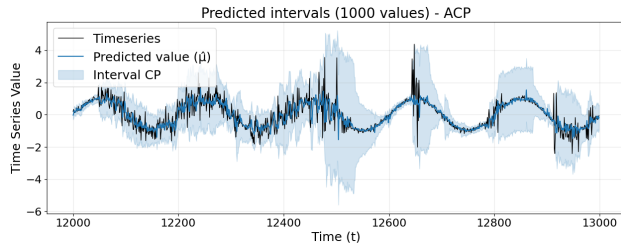


Figure 1: Interval inflation in ACP after a burst of large errors, with a sliding calibration window of size 100 and  $\gamma$  tuned to minimize mean interval width. Scores collected during the disturbance keep the intervals wide well beyond it.

as an online regime detector. Instead of one calibration window, the method maintains one weighted calibration set *per hidden state*: each new score is inserted into the sets of all states whose posterior probability exceeds a threshold  $\tau$ , weighted by that probability. At prediction time, the most likely current state is identified and the interval is built from a weighted empirical quantile of that state’s calibration set, at a level  $1 - \alpha_t^{(k)}$  maintained by a per-state ACP recursion. Scores from different regimes therefore no longer contaminate one another, while the ACP feedback loop still corrects any residual miscoverage within each regime.

## Contributions.

- We introduce regime-aware ACP (RA-ACP), an online conformal prediction method that conditions calibration on latent regimes discovered from the forecaster’s error distribution rather than from the raw series - via a lightweight HMM fitted to log-scaled errors - using posterior-weighted calibration sets and a weighted empirical quantile (Section 4).
- We prove that RA-ACP inherits the asymptotic coverage guarantee of ACP: under a weak-exogeneity condition, the empirical miscoverage converges to the target  $\alpha$  at rate  $O(K/(\gamma T))$  for  $K$  states, both marginally and within every regime that is visited infinitely often (Section 5).
- We evaluate RA-ACP on a synthetic regime-switching benchmark, household electric power consumption, and ECG recordings, at coverage targets 0.90, 0.95, and 0.99, against ACP, EnbPI, and SPCI. RA-ACP always stays within 0.01 of nominal coverage while reducing median interval width by up to 71% and mean width by up to 34% relative to ACP; EnbPI and SPCI attain narrow

intervals only at the price of coverage violations on the real data sets (Section 6).

## 2 RELATED WORK

**Conformal prediction.** Conformal prediction originates with Vovk et al. (2005); Angelopoulos and Bates (2023) give a modern introduction. Split CP calibrates a quantile of nonconformity scores on held-out data and guarantees marginal coverage for exchangeable data (Lei et al., 2018), with numerous refinements such as conformalized quantile regression (Romano et al., 2019). Weighted variants extend validity beyond exchangeability: Tibshirani et al. (2019) reweight scores under covariate shift, and Barber et al. (2023) bound the coverage gap of weighted CP under general distribution drift. Our weighted empirical quantile is in the same spirit, with weights supplied by the posterior of a latent-state model.

**Conformal prediction for time series.** ACP (Gibbs and Candès, 2021) updates the miscoverage level by online gradient descent on the pinball loss and is valid for arbitrary, even adversarial, sequences. Follow-up work adapts its step size by expert aggregation (Zaffran et al., 2022) or handles arbitrary distribution shifts with dynamic regret guarantees (Gibbs and Candès, 2024). EnbPI (Xu and Xie, 2021, 2023a) builds intervals from leave-one-out ensemble residuals in a sliding window, and SPCI (Xu and Xie, 2023b) replaces the empirical quantile with a quantile regression on lagged residuals. HopCPT (Auer et al., 2023) is closest in motivation to our work: it reweights past scores by regime similarity retrieved with a modern Hopfield network. Unlike HopCPT, our method uses a lightweight generative HMM, keeps the online ACP correction within each regime, and retains ACP-style distribution-free coverage guarantees.

**Conditional and Mondrian CP.** Mondrian conformal prediction (Vovk, 2012) partitions observations into categories and calibrates within each, guaranteeing category-conditional coverage for a *fixed, observable* taxonomy. RA-ACP can be viewed as an online Mondrian ACP whose categories are *latent* regimes learned from forecasting errors, with soft assignments handled by posterior weighting.

**CP for TS with change points.** Closest to our work, (Sun and Yu, 2025) propose CPTC, which pairs a switching state-space model with per-state calibration sets and per-state ACI recursions, aggregating state-specific intervals via a weighted level set. Like RA-ACP, it decomposes the global ACI recursion into  $K$  state-conditional recursions and inherits an asymp-

otic validity guarantee. The two methods differ in what the state model conditions on and how intervals are combined. CPTC’s switching dynamical system is fitted to raw observations, requires a state-conditioned forecaster and hours of GPU training per dataset, and has a comparatively rigid transition structure that tracks only coarse, slow-changing regimes. RA-ACP instead detects regimes directly in the forecaster’s error signal with a lightweight HMM trained in seconds, decoupling detection from forecasting and re-clustering far more responsively to brief bursts of miscalibration; unlike RED-SDS, this HMM can also be updated online via incremental EM, letting the regime structure itself adapt as the series evolves rather than remaining fixed after an offline fit. CPTC also aggregates by taking the union of the most probable states’ intervals, which can produce a disjoint prediction region when probability mass is split across states with distant conditional means.

**Regime switching and HMMs.** Markov-switching models are a classical device for capturing abrupt changes in time series dynamics (Hamilton, 1989). HMMs (Baum and Petrie, 1966; Rabiner, 1989) provide efficient filtering of the latent state via the forward-backward recursions and can be estimated by Baum-Welch (Baum et al., 1970) or online EM (Cappé, 2011). We use a Gaussian HMM on log-scaled errors as a simple, fast regime detector.

### 3 PRELIMINARIES

We observe a univariate time series  $y_1, y_2, \dots$  and form covariates  $x_t$  from recent lags. The observation indices are split chronologically into a training set  $\mathcal{D}_{\text{Tr}}$ , a calibration set  $\mathcal{D}_{\text{Cal}}$ , and a test set  $\mathcal{D}_{\text{Te}}$ ; a point forecaster  $\hat{\mu}(\cdot)$  is fitted on  $\mathcal{D}_{\text{Tr}}$ .

#### 3.1 Split Conformal Prediction

Split CP (Vovk et al., 2005; Lei et al., 2018) computes nonconformity scores on the calibration set,

$$\mathcal{C} = \{s_i = |y_i - \hat{\mu}(x_i)| : i \in \mathcal{D}_{\text{Cal}}\}, \quad (1)$$

and lets  $\hat{Q}_{1-\alpha}(\mathcal{C})$  be the  $\lceil (1-\alpha)(1+1/n) \rceil$ -th smallest score, with  $n = |\mathcal{C}|$ . The prediction interval

$$\hat{C}_\alpha(x_t) = [\hat{\mu}(x_t) \pm \hat{Q}_{1-\alpha}(\mathcal{C})] \quad (2)$$

covers  $y_t$  with probability at least  $1 - \alpha$  whenever the data are exchangeable—an assumption that temporal dependence and distribution shift routinely break.

#### 3.2 Adaptive Conformal Prediction

ACP (Gibbs and Candès, 2021) replaces the fixed level  $\alpha$  with a working level  $\alpha_t$ , updated after each observation:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = \mathbf{1}\{y_t \notin \hat{C}_{\alpha_t}(x_t)\}, \quad (3)$$

where  $\gamma > 0$  is a step size. A miss ( $\text{err}_t = 1$ ) increases  $\alpha_t$ ’s complement—the next interval widens; a hit narrows it. The calibration set is a sliding window of the  $N$  most recent scores. The empirical miscoverage of ACP converges to  $\alpha$  for *every* sequence, at rate  $O(1/(\gamma T))$  (Gibbs and Candès, 2021). Its weakness is the shared window: a burst of large scores dominates  $\hat{Q}_{1-\alpha_t}$  for up to  $N$  subsequent steps regardless of the current regime, and the recursion (3) can only compensate slowly (Figure 1).

#### 3.3 Hidden Markov Models

An HMM  $\lambda = (A, B, \pi)$  posits a latent Markov chain  $z_t \in \{1, \dots, K\}$  with transition matrix  $A$  and initial distribution  $\pi$ , emitting observations through state-conditional densities  $B$  (Rabiner, 1989). Given a window of recent observations, the forward recursion yields the filtering posterior  $\mathbb{P}(z_t = k \mid v_{t-h+1:t})$  over states; parameters are estimated with Baum-Welch (Baum et al., 1970). We only require these posteriors, computed over a short window of length  $h$ , as the interface between the HMM and the conformal layer.

### 4 REGIME-AWARE ADAPTIVE CONFORMAL PREDICTION

The idea of RA-ACP is to maintain several calibration sets and choose the active one according to the current performance regime of the forecasting model. An HMM is fitted (on training data) to a transformation  $v_t$  of the forecasting errors, and its filtering posterior

$$w_t^{(k)} = \mathbb{P}(z_t = k \mid v_{t-h+1:t}), \quad k = 1, \dots, K, \quad (4)$$

drives both the update of the calibration sets and the choice of the set used for prediction. Figure 2 summarizes the architecture. Throughout,  $\tau \in (0, 1)$  is a probability threshold,  $N$  the capacity of each calibration set, and  $\mathcal{C}^{(k)}$  the weighted calibration set of state  $k$ .

#### 4.1 State-Specific Calibration Sets

When the true value  $y_t$  becomes available, the nonconformity score  $s_t = |y_t - \hat{\mu}(x_t)|$  is computed together with the posterior weights (4). For every state  $k$  with  $w_t^{(k)} \geq \tau$ , the pair  $(s_t, w_t^{(k)})$  is appended to  $\mathcal{C}^{(k)}$ ; if

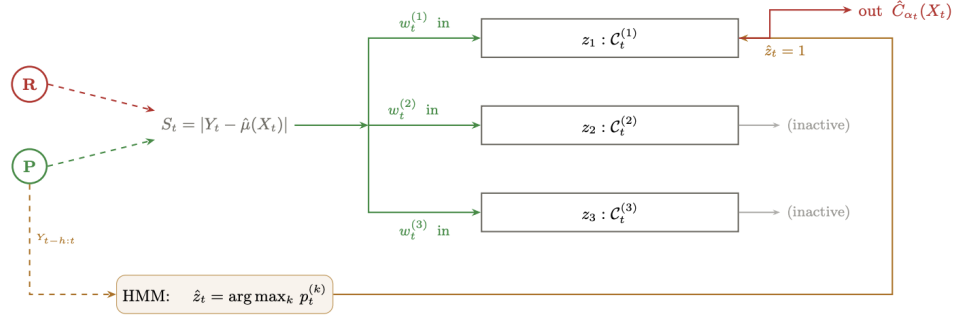


Figure 2: Regime-aware ACP. The HMM assigns each forecasting error a posterior over latent regimes; scores enter the calibration sets of all sufficiently probable states (weighted by the posterior), and the interval at time  $t$  is a weighted quantile of the most likely state’s set. R and P denote the real and predicted value.

$|\mathcal{C}^{(k)}| > N$ , the oldest element is dropped. Thresholding by  $\tau$  prevents low-confidence states from accumulating noisy observations, while the weights record how strongly each score belonged to the regime. If no state reaches the threshold, the score is assigned to the most likely state, so that no observation is discarded.

## 4.2 Online Prediction

At time  $t$ , the active state is the maximum a posteriori state given the data available *before* observing  $y_t$ :

$$\hat{z}_t = \arg \max_k \mathbb{P}(z_t = k \mid v_{t-h:t-1}). \quad (5)$$

Let  $\mathcal{C}^{(\hat{z}_t)} = \{(s_1, w_1), \dots, (s_n, w_n)\}$  and let  $\tilde{w}_i = w_i / \sum_{j=1}^n w_j$  be the normalized weights. Writing  $s_{(1)} \leq \dots \leq s_{(n)}$  for the sorted scores with correspondingly permuted weights  $\tilde{w}_{(i)}$ , the *weighted empirical quantile* at level  $\beta$  is

$$\hat{Q}_\beta(\mathcal{C}^{(\hat{z}_t)}) = s_{(k^*)}, \quad k^* = \min \left\{ k : \sum_{i=1}^k \tilde{w}_{(i)} \geq \beta \right\}. \quad (6)$$

The prediction interval and the per-state ACP recursion are

$$\hat{C}_t(x_t) = \left[ \hat{\mu}(x_t) \pm \hat{Q}_{1-\alpha_t^{(\hat{z}_t)}}(\mathcal{C}^{(\hat{z}_t)}) \right], \quad (7)$$

$$\alpha_{t+1}^{(\hat{z}_t)} = \alpha_t^{(\hat{z}_t)} + \gamma(\alpha - \text{err}_t), \quad (8)$$

where  $\text{err}_t = \mathbf{1}\{y_t \notin \hat{C}_t(x_t)\}$ , with  $\alpha_{t+1}^{(k)} = \alpha_t^{(k)}$  for all inactive states  $k \neq \hat{z}_t$  and  $\alpha_1^{(k)} = \alpha$  for all  $k$ . Each hidden state therefore runs its own ACP feedback loop on the sub-sequence of time steps in which it is active: the interval adapts both to the regime identified by the HMM (through the calibration set and its weights) and to the residual miscoverage within that regime (through  $\alpha_t^{(k)}$ ). Algorithm 1 summarizes the procedure.

---

### Algorithm 1 Regime-aware ACP (RA-ACP)

---

**Input:** sets  $\mathcal{D}_{\text{Tr}}$ ,  $\mathcal{D}_{\text{Cal}}$ , test stream  $\mathcal{D}_{\text{Te}}$

**Parameters:** target level  $\alpha$ , step size  $\gamma$ , states  $K$ , threshold  $\tau$ , window  $h$ , capacity  $N$

- 1: Train forecaster  $\hat{\mu}(\cdot)$  and HMM on  $\mathcal{D}_{\text{Tr}}$
  - 2:  $\alpha_1^{(k)} \leftarrow \alpha$ ;  $\mathcal{C}^{(k)} \leftarrow \emptyset$  **for**  $k = 1, \dots, K$
  - 3: **for**  $i \in \mathcal{D}_{\text{Cal}}$  **and then**  $t \in \mathcal{D}_{\text{Te}}$  (after line 14) **do**
  - 4:  $s \leftarrow |y - \hat{\mu}(x)|$ ; compute  $w^{(k)}$  by (4)
  - 5: **for**  $k = 1, \dots, K$  with  $w^{(k)} \geq \tau$  **do**
  - 6: append  $(s, w^{(k)})$  to  $\mathcal{C}^{(k)}$ ; drop oldest if  $|\mathcal{C}^{(k)}| > N$
  - 7: **end for**
  - 8: **end for**
  - 9: **for**  $t \in \mathcal{D}_{\text{Te}}$  **do**
  - 10:  $\hat{z}_t \leftarrow \arg \max_k \mathbb{P}(z_t = k \mid v_{t-h:t-1})$
  - 11:  $\hat{Q} \leftarrow \hat{Q}_{1-\alpha_t^{(\hat{z}_t)}}(\mathcal{C}^{(\hat{z}_t)})$  by (6)
  - 12: predict  $\hat{C}_t(x_t) = [\hat{\mu}(x_t) \pm \hat{Q}]$
  - 13: observe  $y_t$ ;  $\text{err}_t \leftarrow \mathbf{1}\{y_t \notin \hat{C}_t(x_t)\}$
  - 14:  $\alpha_{t+1}^{(\hat{z}_t)} \leftarrow \alpha_t^{(\hat{z}_t)} + \gamma(\alpha - \text{err}_t)$
  - 15: update calibration sets with  $(y_t, x_t)$  as in lines 4–7
  - 16: **end for**
- 

In practice,  $\hat{Q}_\beta$  is computed with linear interpolation between adjacent order statistics, and levels  $\beta$  outside  $[0, 1]$  (possible since  $\alpha_t^{(k)}$  may exit  $[0, 1]$  by at most  $\gamma$ ) are clipped to the smallest and largest score in the set, which keeps all intervals finite.

## 4.3 Instantiation: Gaussian HMM on Log Errors

The regime detector must track the *performance* of the forecaster rather than the raw signal, so the HMM is fitted to the log-scaled absolute errors  $v_t = \log |y_t - \hat{\mu}(x_t)|$  with Gaussian emissions per state. Modeling

the log-error as Gaussian corresponds to a log-normal distribution of the raw error magnitude—a natural choice for strictly positive, right-skewed residuals—and matches the Gaussian-mixture emissions supported by standard HMM implementations. States then correspond to error-magnitude regimes (e.g., calm, moderate, and turbulent), which is exactly the structure the calibration sets should separate.

#### 4.4 Assignment and the role of $\tau$

Hard assignment - routing each score to the single most likely state discards information whenever more than one state has meaningful posterior support: a score might genuinely belong to few plausible regimes, and committing it to only one throws away evidence the other state could have used. RA-ACP instead let a score enter every state whose posterior exceeds  $\tau$ , weighted by that state's probability, so an observation with support across several states contributes to each of their calibration sets rather than being assigned to just one. Without such a threshold, every state with any nonzero posterior -however small -would receive an entry at every single time step, since soft-weighting alone never rules a state out completely. Over enough iterations, states that are rarely or never truly active would still accumulate large calibration sets built almost entirely out of noise, scores that happened to carry a tiny sliver of posterior mass rather than any genuine membership in that regime.

The weight itself matters beyond the threshold: two states that both clear  $\tau$  need not be equally likely, and treating their scores as equally informative would let a barely-admissible state distort the quantile as much as a strongly supported one; weighting by  $w_t^k$  lets the calibration set reflect not just which states are plausible, but how plausible each one is.

This soft, thresholded assignment governs which scores populate each state's calibration set, but does not itself affect the coverage guarantee below, which depends only on the per-state ACI recursion operating on active calibration set  $C^k$  at each step. In particular, the threshold and weights depend only on the HMM posterior and not on the values  $\alpha_t^k$  themselves.

#### 4.5 Levels outside $[0, 1]$

In the original ACI, a level  $1 - \alpha_t \notin [0, 1]$  yields an infinite (or empty) set (Gibbs and Candès, 2021), the convention under which Proposition 2 is stated. In our experiments we instead clip such levels to the smallest and largest score in the calibration set. This is common in practice: infinite intervals are uninformative and make mean width undefined. Angelopoulos et al. (2023) likewise evaluate a variant of ACI clipped to the

largest score seen so far. As they note, such clipping is a heuristic and the guarantee can be affected only at steps where clipping is actually triggered.

## 5 COVERAGE GUARANTEES

The guarantee of ACP is distribution-free: for any sequence,  $\alpha_t \in [-\gamma, 1 + \gamma]$  for all  $t$ , and the empirical miscoverage after  $T$  steps deviates from  $\alpha$  by at most  $(\max\{\alpha_1, 1 - \alpha_1\} + \gamma)/(\gamma T)$  (Gibbs and Candès, 2021, Lemma 4.1 and Proposition 4.1). Because RA-ACP runs one such recursion per state, the guarantee lifts to the regime-aware setting. We require that the regime process does not react to the calibration parameters:

**Assumption 1** (Weak exogeneity). *For every state  $k$  and every  $t$ , the active state satisfies  $\hat{z}_t \perp \alpha_t^{(k)} \mid \hat{z}_{t-1}$ ; that is, given the previous state, regime switches are independent of the current values of the per-state miscoverage parameters.*

**Proposition 2.** *Let the active-state process  $\{\hat{z}_t\}_{t \in \mathbb{N}}$  take values in  $\{1, \dots, K\}$  and let Assumption 1 hold. Then, for every realization and every  $T \in \mathbb{N}$ ,*

$$\left| \frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right| \leq \frac{K(\max\{\alpha, 1 - \alpha\} + \gamma)}{\gamma T} \xrightarrow{T \rightarrow \infty} 0. \quad (9)$$

*Proof.* Fix a state  $k$  and let  $T_k = |\{t \leq T : \hat{z}_t = k\}|$ . On the sub-sequence of steps in which state  $k$  is active, the parameter  $\alpha_t^{(k)}$  evolves exactly by the ACP recursion (8) with initialization  $\alpha_1^{(k)} = \alpha$ , and by Assumption 1 the sub-sequence is not chosen adversarially with respect to  $\alpha_t^{(k)}$ . Proposition 4.1 of Gibbs and Candès (2021) applied to this sub-sequence gives

$$\left| \sum_{t: \hat{z}_t = k} \text{err}_t - T_k \alpha \right| \leq \frac{\max\{\alpha, 1 - \alpha\} + \gamma}{\gamma},$$

with a right-hand side independent of  $T_k$ . Every step is assigned to exactly one active state, so  $\sum_{k=1}^K T_k = T$ , and summing the bound over  $k$  yields

$$\left| \sum_{t=1}^T \text{err}_t - T\alpha \right| \leq \frac{K(\max\{\alpha, 1 - \alpha\} + \gamma)}{\gamma}.$$

Dividing by  $T$  gives the claim.  $\square$

*Remark 3.* The same argument applied to a single sub-sequence shows that for every state  $k$  visited infinitely often,  $\frac{1}{T_k} \sum_{t: \hat{z}_t = k} \text{err}_t \rightarrow \alpha$ . Coverage is therefore preserved not only marginally but within every active regime—a conditional guarantee that a single global ACP recursion does not provide.

The price of regime-awareness is a factor of  $K$  in the finite-sample bound, which is immaterial for the small numbers of states ( $K \leq 4$ ) selected in our experiments.

## 6 EXPERIMENTS

### 6.1 Setup

**Protocol.** Each data set is split chronologically into training (4,000 samples), calibration (2,000), validation (4,000), and test (4,000) segments. The point forecaster for all methods is a random forest (Breiman, 2001) in autoregressive mode,  $\hat{y}_t = \text{RF}(y_{t-1}, \dots, y_{t-4})$ . Experiments are run at coverage targets  $1 - \alpha \in \{0.90, 0.95, 0.99\}$ ; for the windowed methods (ACP, RA-ACP) we report calibration-window capacities  $N \in \{100, 300, 600\}$ , with  $N = 600$  in the main text and the remaining settings in Appendix A.

**Baselines.** *ACP* (Gibbs and Candès, 2021) with a sliding calibration window and the update (3). *EnbPI* (Xu and Xie, 2021) trains an ensemble of bootstrap models and forms intervals from a sliding window of leave-one-out residuals. *SPCI* (Xu and Xie, 2023b) fits a quantile random forest on lagged residuals to predict the next conformal quantile. EnbPI and SPCI use the concatenated training and calibration segments, as they do not need a separate calibration set.

**Hyperparameters.** All method hyperparameters (for RA-ACP:  $\gamma$ , the number of HMM states  $K$ , the threshold  $\tau$ , and the HMM window  $h$ ) are tuned with Optuna’s TPE sampler (Akiba et al., 2019), minimizing mean interval width on the validation segment; coverage is then reported on the untouched test segment. TPE is preferred over surrogate-based optimizers because the mapping from the continuous  $\gamma$  to the selected discrete score is piecewise constant and poorly captured by smooth surrogates. Selected values are listed in Appendix B; notably, the optimal  $\gamma$  for RA-ACP is often at the lower search bound, indicating that most of the adaptation is carried by regime switching rather than by the  $\alpha_t^{(k)}$  recursion.

**Metrics.** We report empirical coverage  $\frac{1}{T} \sum_t \mathbf{1}\{y_t \in \hat{C}_t\}$ , mean interval width, and median interval width on the test segment. The median complements the mean because a small number of extremely wide intervals—precisely the failure mode of regime-blind calibration—can dominate the mean while leaving typical intervals unchanged.

### 6.2 Data Sets

We test our method on three different datasets:

- *Synthetic*: a sinusoidal signal with additive Gaussian noise whose variance switches among three regimes (low, medium, high volatility) according

to a Markov chain; the transition matrix is given in Appendix D.

- *Power*: household electric power consumption recorded at one-minute resolution in a French household between 2006 and 2010 (Hebrail and Berard, 2012); we forecast the active-power series.
- *Heartbeat*: ECG recordings from the MIT-BIH Arrhythmia Database (Moody and Mark, 2001; Goldberger et al., 2000); we use the MLII-lead potential, a standard benchmark in biomedical signal processing.

Overview plots of all three series are shown in Appendix D.

### 6.3 Results

Table 1 reports test-set coverage and interval widths at calibration capacity  $N = 600$ ; Appendix A contains all capacities.

**Coverage.** ACP and RA-ACP stay within 0.01 of the nominal level in *every* configuration, across data sets, targets, and calibration capacities. The baselines do not: EnbPI undercovers at the 0.90 target on both real data sets (0.869 on power, 0.866 on heartbeat), and SPCI fails badly on the heartbeat data, reaching 0.787, 0.848, and 0.887 at the 0.90, 0.95, and 0.99 targets. Interval widths of methods that miss the target must be interpreted accordingly.

**Interval widths.** Among coverage-valid methods, RA-ACP produces the narrowest intervals in nearly every configuration. Relative to ACP at  $N = 600$ , it reduces median width by up to 62% and mean width by up to 24% (heartbeat and synthetic at the 0.99 target); across all calibration capacities the reductions reach 71.3% in median and 34.1% in mean width (heartbeat,  $N = 300$ , target 0.99; Appendix A). The gains grow with the coverage target, where regime-blind calibration is most damaged by extreme scores. EnbPI and SPCI attain competitive *mean* widths at low targets, but their *median* widths exceed ACP’s by up to 39.6% and 25.7% on the power data—their intervals are uniformly moderate rather than adapted—and EnbPI’s mean width deteriorates at the 0.99 target (+13.7% over ACP on synthetic data). SPCI’s narrow intervals on the heartbeat data come with the coverage violations noted above.

**Interval dynamics.** Figure 3e shows interval width over time on the heartbeat data. RA-ACP exhibits the intended behavior: widths switch abruptly between levels as the HMM detects regime changes, staying narrow in calm segments and widening only inside

| Data set  | $1-\alpha$ | ACP   |              |       | RA-ACP (ours) |              |              | EnbPI              |       |       | SPCI               |             |       |
|-----------|------------|-------|--------------|-------|---------------|--------------|--------------|--------------------|-------|-------|--------------------|-------------|-------|
|           |            | Cov.  | Mean         | Med.  | Cov.          | Mean         | Med.         | Cov.               | Mean  | Med.  | Cov.               | Mean        | Med.  |
| Synthetic | 0.90       | 0.899 | 1.90         | 1.70  | 0.891         | <b>1.69</b>  | <b>1.46</b>  | 0.891              | 1.77  | 1.76  | 0.891              | 1.84        | 1.74  |
|           | 0.95       | 0.949 | 2.92         | 2.17  | 0.948         | 2.56         | <b>2.11</b>  | 0.945              | 2.43  | 2.43  | 0.942              | <b>2.35</b> | 2.28  |
|           | 0.99       | 0.990 | 5.25         | 4.87  | 0.989         | <b>3.98</b>  | <b>3.14</b>  | 0.989              | 5.97  | 6.17  | 0.980              | 4.19        | 4.59  |
| Power     | 0.90       | 0.895 | 1.22         | 0.60  | 0.893         | <b>1.07</b>  | <b>0.53</b>  | 0.869 <sup>†</sup> | 0.82  | 0.77  | 0.887 <sup>†</sup> | 0.83        | 0.68  |
|           | 0.95       | 0.948 | 1.81         | 1.01  | 0.948         | <b>1.58</b>  | <b>0.83</b>  | 0.934 <sup>†</sup> | 1.48  | 1.41  | 0.937 <sup>†</sup> | 1.42        | 1.27  |
|           | 0.99       | 0.989 | 3.18         | 3.38  | 0.982         | <b>2.54</b>  | <b>2.32</b>  | 0.985              | 3.77  | 3.59  | 0.982              | 2.71        | 2.73  |
| Heartbeat | 0.90       | 0.900 | <b>21.62</b> | 18.16 | 0.902         | 21.69        | <b>14.56</b> | 0.866 <sup>†</sup> | 17.78 | 17.94 | 0.787 <sup>†</sup> | 14.36       | 14.34 |
|           | 0.95       | 0.950 | 31.55        | 26.00 | 0.954         | <b>31.20</b> | <b>18.34</b> | 0.936 <sup>†</sup> | 27.31 | 27.81 | 0.848 <sup>†</sup> | 19.52       | 19.77 |
|           | 0.99       | 0.991 | 74.50        | 74.80 | 0.992         | <b>57.78</b> | <b>28.22</b> | 0.987              | 67.52 | 67.88 | 0.887 <sup>†</sup> | 34.12       | 34.70 |

Table 1: Test-set results at calibration capacity  $N = 600$ . Cov. is empirical coverage; Mean/Med. are mean and median interval widths. <sup>†</sup> marks coverage deviating from the target by more than 0.01. Best mean and median widths *among methods meeting the target within 0.01* are in bold. Full results for  $N \in \{100, 300\}$  are in Appendix A.

turbulent ones. ACP instead drifts gradually, EnbPI produces an almost constant width that adapts only to long-term trends, and SPCI reacts dynamically but with frequent sharp spikes. The width histograms in Appendix E show the same effect as clearly separated width modes for RA-ACP.

**Computational cost.** RA-ACP costs about twice as much as plain ACP (7.9s vs. 3.9s for a full train-and-predict run on 10,000 samples) and is comparable to EnbPI (7.2s), while SPCI is over  $40\times$  more expensive (341.6s) because it refits quantile regressions online (Appendix C).

## 7 LIMITATIONS

RA-ACP relies on the HMM identifying regimes that are meaningful for the forecaster’s error distribution: a misspecified emission model or a poorly chosen number of states  $K$  degrades the width gains (though, by Proposition 2, not the asymptotic coverage). The guarantee itself is asymptotic and its finite-sample bound weakens by a factor of  $K$ . Finally, our evaluation covers univariate series with a single forecaster class and one run per configuration; multivariate settings, other forecasters, and variance across runs remain to be studied.

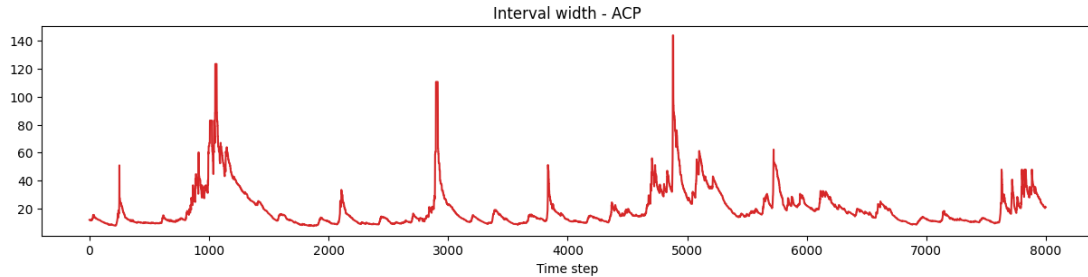
## 8 CONCLUSION

We introduced RA-ACP, a regime-aware extension of adaptive conformal prediction that maintains posterior-weighted, state-specific calibration sets driven by an HMM over log-scaled forecasting errors, and builds intervals from weighted quantiles at per-state adaptive levels. The method provably inherits

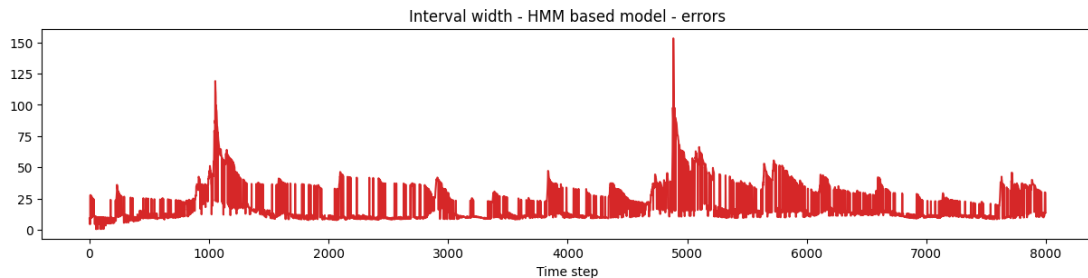
ACP’s distribution-free asymptotic coverage guarantee, marginally and per regime, and empirically delivers substantially narrower intervals than ACP-median reductions of up to 71%-while maintaining nominal coverage where recent baselines (EnbPI, SPCI) violate it. Future work includes adapting  $\gamma$  online, for instance increasing it after consecutive miscoverage events, and re-estimating the HMM online (Cappé, 2011) so that the regime detector itself tracks distribution shift.

### 8.1 Future Work

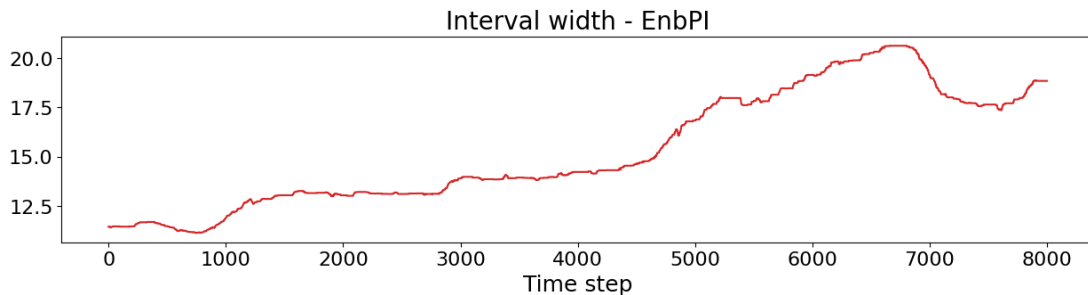
Two natural extensions target the components that are currently fixed by design. First, our HMM is fitted once on the training segment and never updated afterward; replacing this static fit with online EM Cappé (2011) would let the regime detector adapt as the series evolves, rather than relying on a fixed emission model estimated before the distribution shift under study occurs. Second, RA-ACP currently shares a single step size  $\gamma$  across all  $K$  states, even though rare regimes receive feedback far less often than dominant ones and may therefore benefit from faster adaptation; replacing this shared  $\gamma$  with a per-state  $\gamma_k$ , or with a AgACI-style Zaffran et al. (2022) aggregation over step sizes within each state, would let each regime’s recursion adapt at its own natural rate rather than at a rate tuned for the series as a whole. Both extensions preserve the coverage guarantee of Proposition 1, since the proof only requires that each state’s  $\alpha^k$  recursion follows the ACI update with some fixed step size, not that the step size be identical across states or constant in time



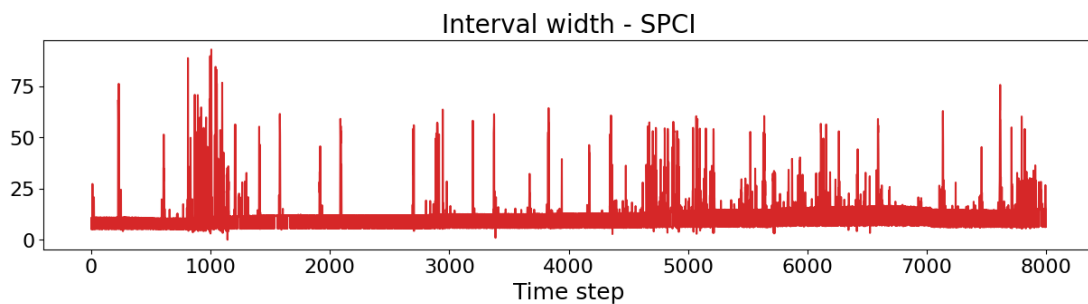
(a) ACP



(b) RA-ACP (ours)



(c) EnbPI



(d) SPCI

(e) Interval width over time on the heartbeat test segment at the 0.90 coverage target. RA-ACP switches between narrow and wide intervals as the HMM tracks ECG regimes; ACP drifts slowly, EnbPI is nearly constant, and SPCI oscillates sharply while undercovering. Analogous plots for all data sets and targets are in Appendix E.

## AI USE STATEMENT

Claude was used to improve the grammar, spelling, and clarity of the text in this paper. The authors verified the accuracy and completeness of the content and confirmed that the use of AI did not introduce any errors or omissions. The authors also verified that

the AI-assisted work did not affect the originality or novelty of the research presented in this paper.

The implementation of the algorithms used to conduct the experiments was written by the authors with the assistance of AI, including suggestions for code implementation, corrections and debugging. The authors reviewed and verified all AI-assisted code.

AI assistance was also used to develop scripts for testing and generating the data presented in tables and figures. The authors reviewed and verified these scripts and confirmed that the reported results and statistics are accurate.

## REFERENCES

- Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2623–2631.
- Angelopoulos, A. N. and Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591.
- Angelopoulos, A. N., Candes, E. J., and Tibshirani, R. J. (2023). Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Auer, A., Gauch, M., Klotz, D., and Hochreiter, S. (2023). Conformal prediction for time series with modern hopfield networks. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845.
- Baum, L. E. and Petrie, T. (1966). Statistical inference for probabilistic functions of finite state Markov chains. *The Annals of Mathematical Statistics*, 37(6):1554–1563.
- Baum, L. E., Petrie, T., Soules, G., and Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Cappé, O. (2011). Online EM algorithm for hidden Markov models. *Journal of Computational and Graphical Statistics*, 20(3):728–749.
- Gibbs, I. and Candès, E. J. (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 1660–1672.
- Gibbs, I. and Candès, E. J. (2024). Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36.
- Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., Mietus, J. E., Moody, G. B., Peng, C.-K., and Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.
- Hebrail, G. and Berard, A. (2012). Individual household electric power consumption. UCI Machine Learning Repository.
- Lei, J., G’Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111.
- Moody, G. B. and Mark, R. G. (2001). The impact of the MIT-BIH arrhythmia database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3):45–50.
- O’Connor, C., Bahloul, M., Rossi, R., Prestwich, S., and Visentin, A. (2025). Conformal prediction for electricity price forecasting in the day-ahead and real-time balancing market. arXiv preprint.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Romano, Y., Patterson, E., and Candès, E. J. (2019). Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 3543–3553.
- Sun, S. and Yu, R. (2025). Conformal prediction for time-series forecasting with change points.
- Tibshirani, R. J., Barber, R. F., Candès, E. J., and Ramdas, A. (2019). Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 2530–2540.
- Vovk, V. (2012). Conditional validity of inductive conformal predictors. In *Proceedings of the Asian Conference on Machine Learning (ACML)*, volume 25 of *Proceedings of Machine Learning Research*, pages 475–490.
- Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer, New York.
- Xu, C. and Xie, Y. (2021). Conformal prediction interval for dynamic time-series. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 11559–11569.
- Xu, C. and Xie, Y. (2023a). Conformal prediction for time series. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10):11575–11587.

- Xu, C. and Xie, Y. (2023b). Sequential predictive conformal inference for time series. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 38707–38727.
- Ye, T., Hijazi, A., and Van Hentenryck, P. (2025). Conformal predictive distributions for order fulfillment time forecasting. arXiv preprint.
- Yuan, S., Li, J., Wang, C., and Zhang, X. (2026). Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems. Preprint.
- Zaffran, M., Féron, O., Goude, Y., Josse, J., and Dieuleveut, A. (2022). Adaptive conformal predictions for time series. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 25834–25866.